Investigation methodologies, collection & acquisition principles and considerations

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Learning outcomes

- Discuss the digital crime scene and the process of evidence collection
- Critically discuss the preservation of evidence
- Develop a critical understanding on the principles of data acquisition
- Critically evaluate the order of volatility in digital forensic investigations

The location where an illegal act took place and comprises the area from which most of the evidence is retrieved

- trained law enforcement personnel
- crime scene investigators (CSIs)
- forensic scientists

Crime Scene

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What Does 'Contact with the Crime Scene' Mean?

- First interaction with a potential scene
- May involve physical and digital environments
- Sets the foundation for the entire investigation
- Early mistakes can be difficult to correct

Created with ChatGPT

Assessing a case

1. Outline the case
2. Determine the case requirements
3. Present a list of case details, e.g. situation, nature, specifics etc.
4. Plan the investigation
5. Secure evidence

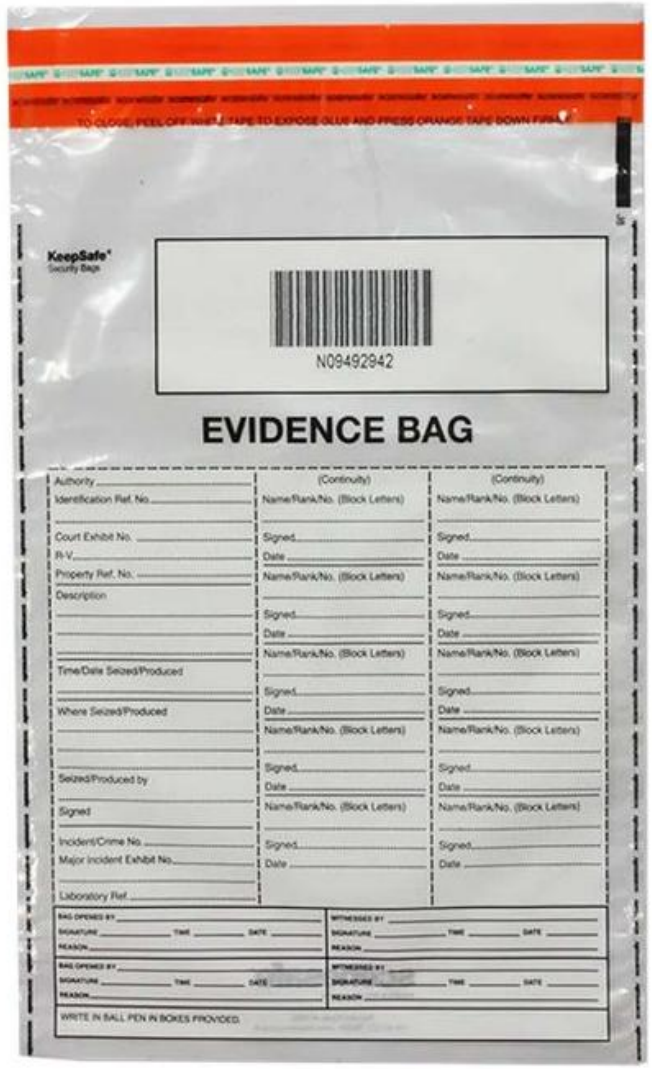
Why First Response Matters

- Evidence can be easily altered or destroyed
- Actions taken early affect admissibility
- Scene management protects integrity
- Professionalism begins at first contact

Search and collection

1. In-depth search and collection of the scene for additional physical evidence.
   - looking for additional media and digital devices at the crime scene.
   - contact people to preserve and obtain access logs
   - the digital crime scene investigation begins now.
   - The computer system itself will be considered physical evidence and therefore collected.
2. Protect the evidence in collection

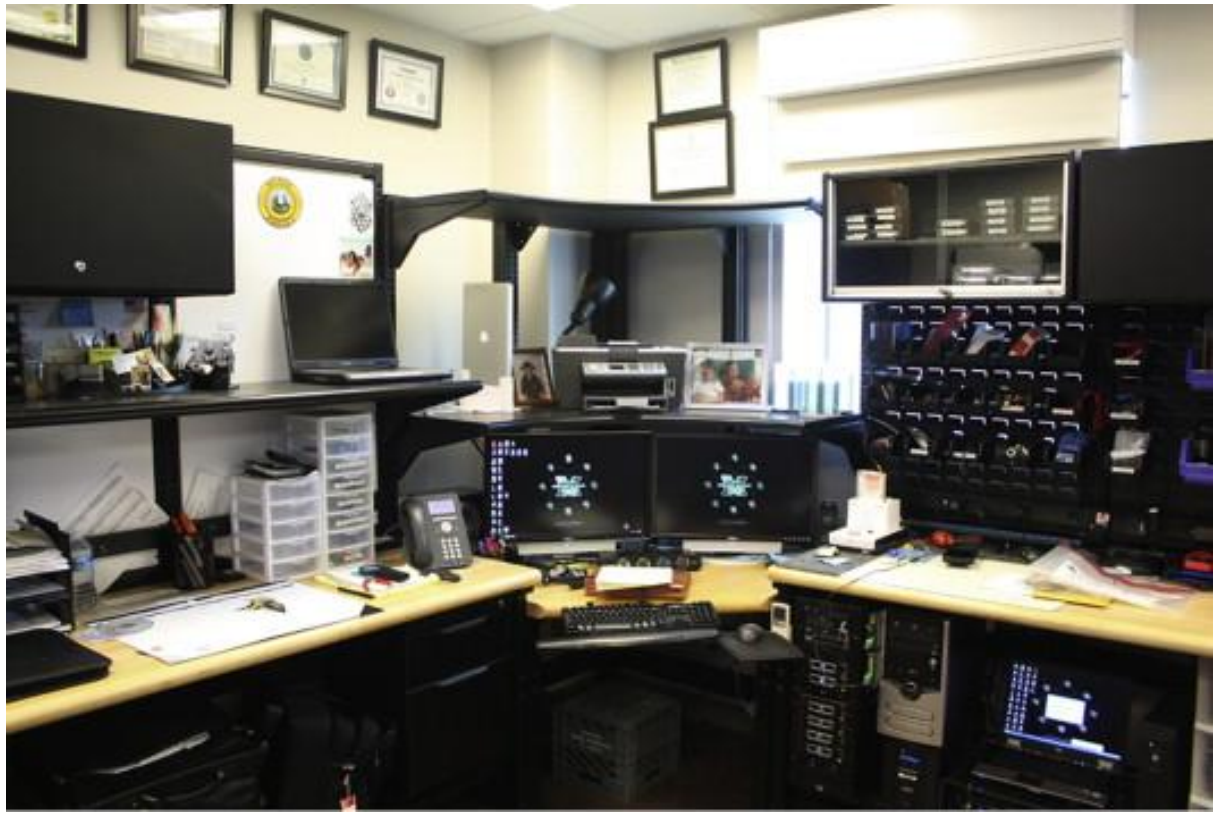
You want to start collecting ASAP!



Bringing evidence in the lab

Maintain the integrity of digital evidence in the lab

- Verify packaging is sealed and intact
- Check labels: case ID, date, handler, exhibit number
- Confirm chain of custody documentation
- Log evidence into secure storage system
- Store in appropriate conditions (locked, access-controlled)
- Do not examine evidence before formal acceptance
- Any issues (damage, broken seals) must be documented immediately



Collection

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Pre-planned Operation

- Corporate internal investigation
- Court issued warrant/ operation
- Appointed by court as an independent investigator

Prepare!!!

- Who's in charge?
- What is your role?
- What is everyone else's role?
- What is expected of you?
- Limitations?
- Emergency plans?

Ask for details!

- Time and date
- Name and address
- Intelligence available
  - risk factors, number of computers / servers
- Specialist knowledge
  - E.g. encryption, memory acquisition

Order of Evidence Collection

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01 Find the evidence

- Figure out where the evidence you are seeking is being stored on the system.

02 Determine data relevance

- Decide what parts of the data you collect are relevant to the case at hand.

03 Rank volatility

- Items that are most likely to degrade or become unusable should be collected first.

04 Eliminate outside interference

- As only unaltered data may be entered as evidence; utmost care must be taken to prevent any possible outside contamination.

05 Collect the evidence

- As you gather evidence, continue to examine the items you've already collected, as new pieces you collect may influence what you consider worthy information.

06 Document everything

- Your method of collecting the evidence you present may be called into question later, so be sure to maintain a record of everything that you've done in the collection process.

Collecting & Preserving Evidence

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Understand volatility of evidence	Some evidence, such as data in the computer's RAM, only exists while the computer is powered on.
Create safe environment	1. Conduct a forensic investigation without booting the computer via the hard drive. 2. Turning on the computer can destroy potential evidence.
Build a forensic toolkit	Essential to any computer investigation, toolkits come in two types; those you assemble yourself and pre-fabricated ones that you download or purchase as a suite from any one of many forensic software vendors.
Follow the chain-of-custody	The chain-of-custody is a record of evidence handling from the time of seizure to the time evidence is presented in a court of law.
Ensure the admissibility of evidence collected via authentication	Before a computer record can be used as evidence, it must first be proven to be authentic.

Evidence retention

- Maintain the chain of custody for digital evidence
- Restrict access to lab and evidence storage area
- Lab should have a sign-in/ out register for all visitors
- Maintain logs for a period based on legal requirements
- You might need to retain evidence indefinitely
- Consider compliance standards

Bag & Tag

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Aim is to protect against contamination and tampering

- Location, date, time, case, event, seizing agent

- When you decide to seize an exhibit you start the chain of custody
- Consider **photographing** the exhibit in-site, photograph the rear of the computer with cables and tags in place
- Consider **labeling**
  - leads and ports, all the cables,
  - how the computer was configured when you found it
  - Reconstruction of the configuration
- Where possible seal exhibit into a bag
  - time, date and sign the exhibit label
- Until somebody formally accepts the exhibit from you, **<u>it is your responsibility</u>**

The forensic image

Hardware write blocker recommended

Acquisition

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Digital evidence storage formats

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Raw format

- Makes it possible to write bit-stream data to files
- Advantages
  - Fast data transfers
  - Ignores minor data read errors on source drive
  - Most computer forensics tools can read raw format
- Disadvantages
  - Requires as much storage as original disk or data
  - Tools might not collect marginal (bad) sectors

Proprietary formats

- Most forensics tools have their own formats
- Features offered
  - Option to compress or not compress image files
  - Can split an image into smaller segmented files
  - Can integrate metadata into the image file
- Disadvantages
  - Inability to share an image between different tools
  - File size limitation for each segmented volume

The acquisition process

Step 1. Select appropriate tool**,** based on the type of case, the investigator's preference and training.

Step 2. Protect media from possible alteration of data - **Preservation**

Step 3. Image the original media, create the image according to relevant rules and standards

Step 4. Store original media safely, at a safe place with limited access, e.g. an evidence locker.

Step 5. Back-up the image, work on that, e.g. some software does so automatically

Step 6. Verify the image - **Integrity** of the evidence for the representation in a court.

Determine the best method

1. Static acquisitions
2. Live acquisitions

Disk-to-image file	Disk-to-disk file	Logical file
Most common method and offers most flexibility Can make more than one copy Copies are bit-for-bit replications of the original drive Compatible with many commercial forensic tools	When disk-to-image copy is not possible Tools can adjust disk's geometry configuration	Disk-to-disk or disk-to-data file Captures only specific files/ folders of interest to the case Also, sparse acquisition collects fragments of unallocated (deleted) data

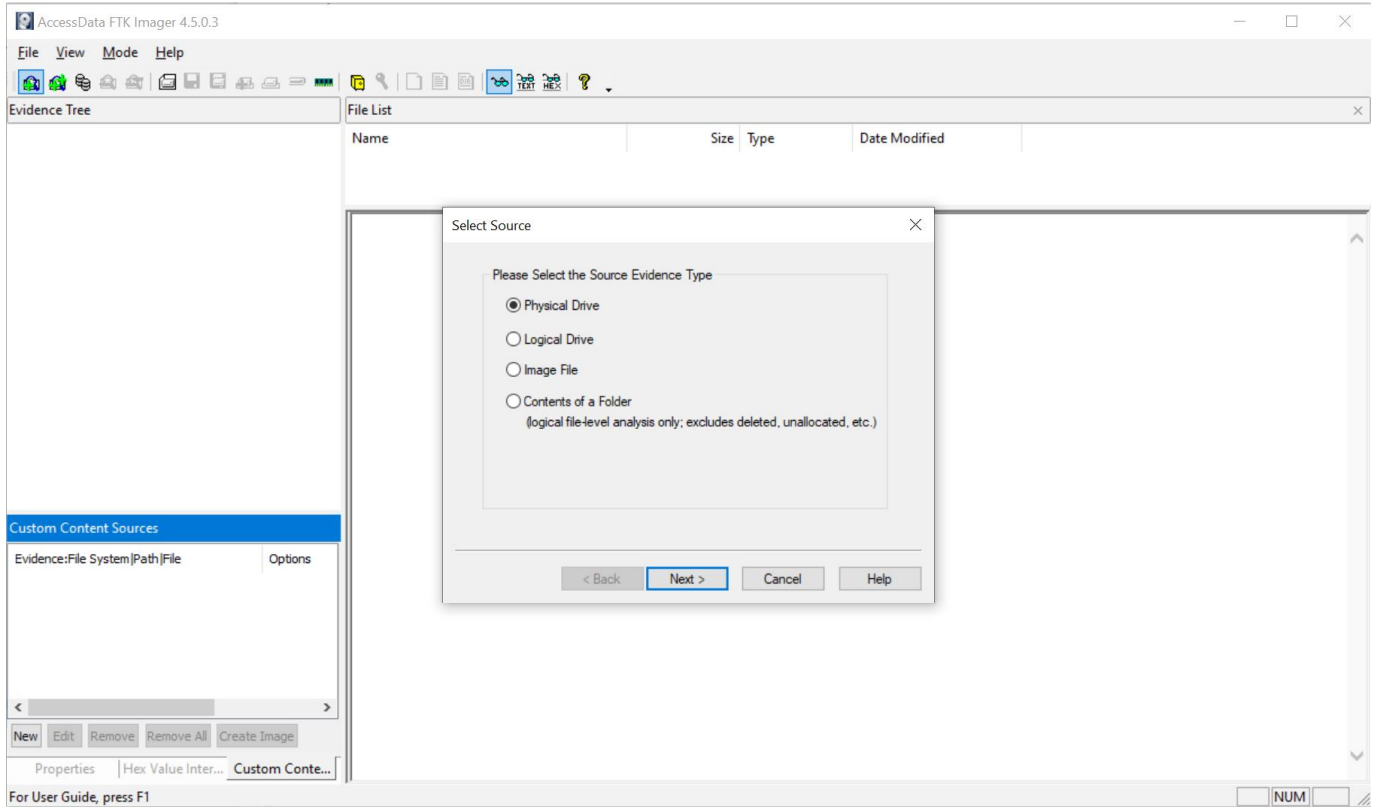
Forensic Acquisition Tools

Advantages

- Make acquiring evidence from a suspect drive more convenient
- Especially when used with hot-swappable devices

Disadvantages

- Must protect acquired data with a well-tested write-blocking hardware device
- Tools can't acquire data from a disk's host protected area
- Some countries haven't accepted the use of write-blocking devices for data acquisitions



Data Validation & Verification

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- Record the cryptographic hashes created by the imaging program

  "A mathematical function that can take in a varying length of data to create a fixed length output that mathematically represents the entire data set in such a way that it is statistically infeasible that two different data sets could ever have the same result."

- The point is to **fingerprint** the pre-image

Message Digest

MD5 algorithm = 128-bit - 32 character hash algorithm

2^{128} likelihood that any 2 files have the same hash value

Secure Hash Algorithm

SHA1 = 160-bit hash algorithm

2^{160} likelihood that any 2 files have the same hash value

anything you look at and anything you present as evidence must be hashed.

1. You can't predict the hash value of a file or device
2. No two hash values can be the same
3. If anything changes in the file or device, the hash value must change

Evidence Preservation

- Forensically wipe image drives
- Preserving original artefacts in a way that is:
  - Reliable, Complete, Accurate, Verifiable
- How?
  - Cryptographic hashing, Checksums, documentation
- If something is wrong, it doesn't *necessarily* mean that the exhibit is lost

Remote acquisitions

- Remote acquisition tools vary in configurations and capabilities
- Drawbacks
  - Antivirus, antispyware, and firewall tools can be configured to ignore remote access programs
  - Suspects could easily install their own security tools that trigger an alarm to notify them of remote access intrusions

- Tools: F-Secure, ProDiscover, EnCase

Exercise

Image the contents of a USB stick that you own to another destination

```
# dd if=/dev/source of=/dev/destination
```

Try:

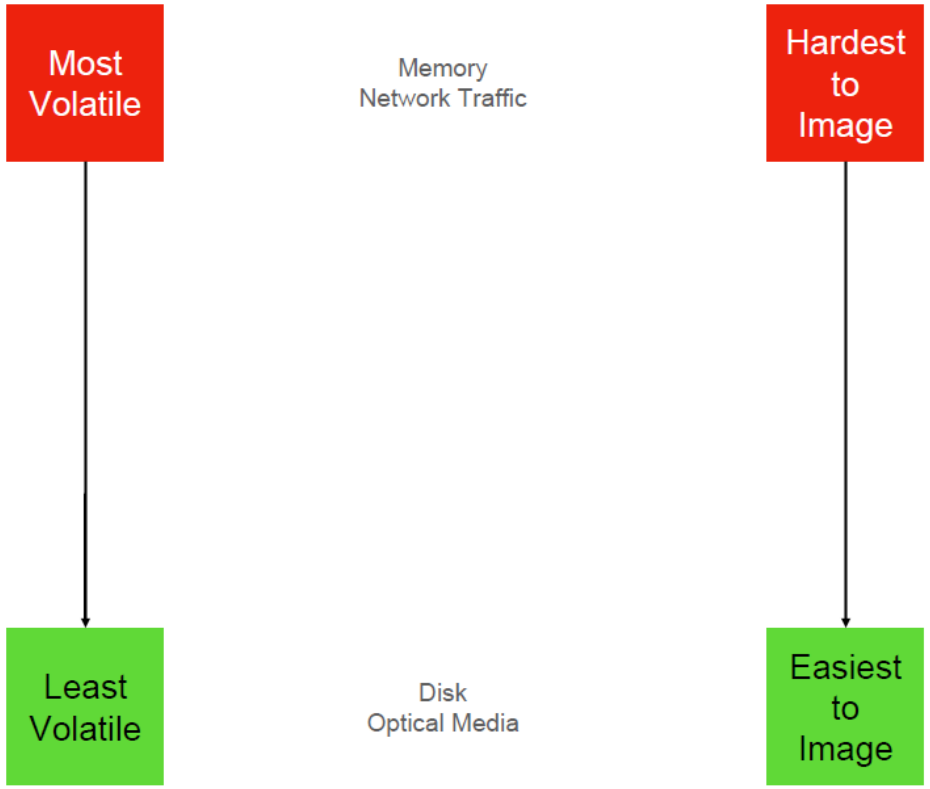
```
# dd if=/dev/source of=/dev/destination bs=128K conv=noerror,sync
```

- `if=/dev/source` — Input device/file
- `of=/dev/destination` — Output device/file
- `bs=128K` — Block size
- `conv=noerror` — Ignore any read errors
- `sync` — Data offsets in sync

ORDER OF VOLATILITY (OOV)

The order of volatility

The more accurate and complete the data, the better and more comprehensive the evaluation can be.



RFC 3227 – Guidelines for Evidence Collection and Archiving

RFC 3227 "Order of Volatility"

Areas traditionally considered "volatile"	1. Registers, cache 2. Routing table, arp cache, process table, kernel statistics, memory 3. Temporary file systems
Areas traditionally considered "non-volatile"	4. Disk 5. Remote logging and monitoring data that is relevant to the system in question 6. Physical configuration, network topology 7. Archival media

Volatile Data Collection

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Memory → System Information → Network Data → Processes and Drivers

Memory

- (nearly) all active content
- System Information
- Active Processes
- Usernames/ Passwords
- Decrypted Programs
- Clipboard content

System Information

- Date & Time
- System Information
- Volume Information
- Logged on users
- Encryption detection
- Autostarts
- File Listing
- Scheduled Tasks

Network data

- Network configuration
- Arp Table
- Routing Cable
- DNS Cache
- Network connections and sockets
- Remote open files

Processes and Drivers

- List of processes
- List of services
- Loaded DLLs
- Drivers

Pitfalls to avoid

- Lacking familiarity with tools, techniques, and protocols prior to an incident
- Practice – Train – Stay current
- Failing to test and validate your tools
- Research – Test – Document
- Using improperly licensed commercial tools
- Not conducting interviews prior to conducting live response
- Running non-trusted tools directly from the subject system
- Not using forensically sound/clean acquisition media
- Not following the order of volatility
- Not following the methodology

Summary

- Protect your safety and health as well as the integrity of the evidence
- Follow guidelines when processing an incident or crime scene
- As you collect digital evidence, guard against physically destroying or contaminating it
- Forensic hash values verify that data or storage media have not been altered
- You must handle all evidence the same way every time you handle it
- After you determine that an incident scene has digital evidence, identify the digital information or artifacts that can be used as evidence

The forensic methodology

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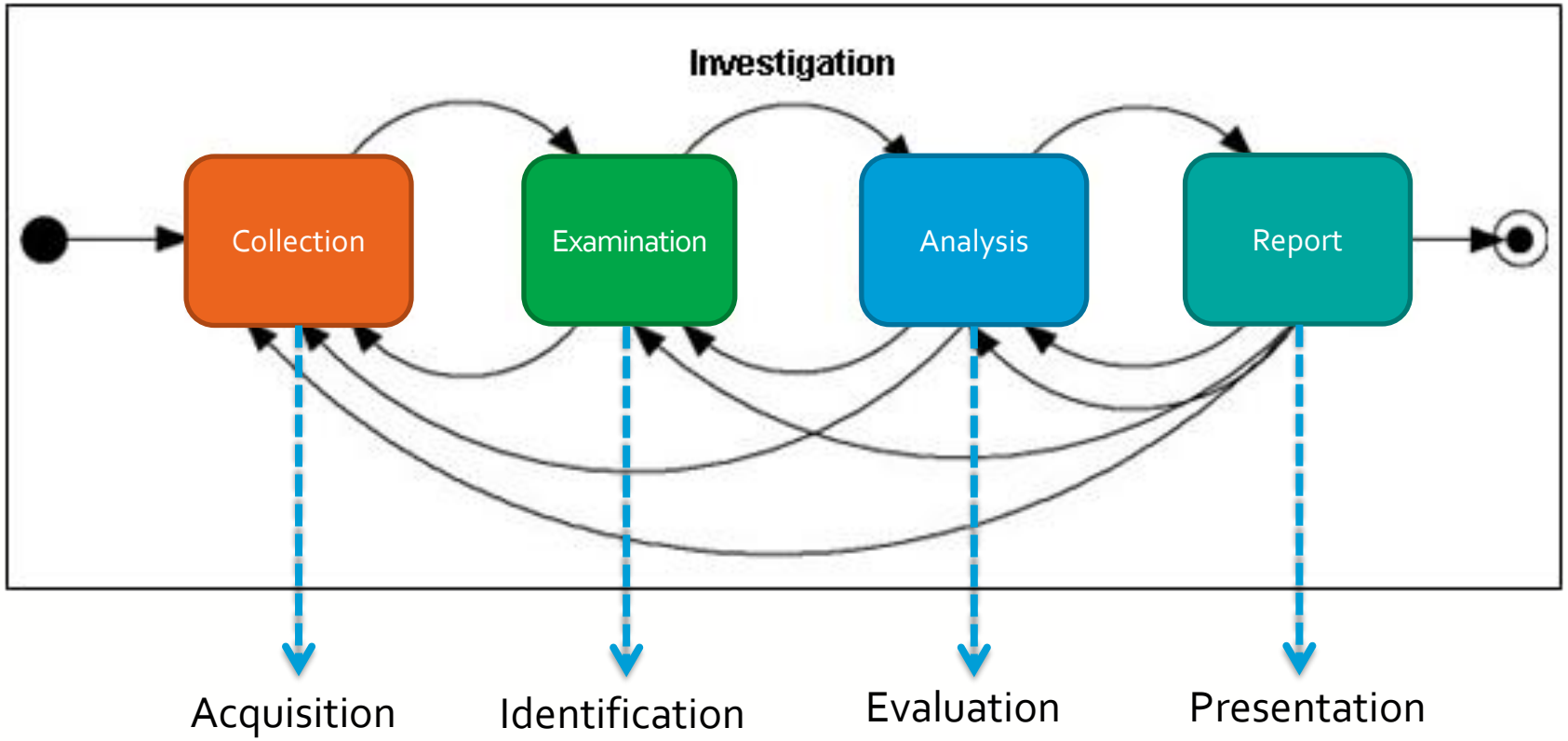
Approaching Digital Forensics Cases

1. Begin a case by creating an investigation plan that defines the:
   - Goal and scope of investigation
   - Materials needed
   - Tasks to perform
2. The approach you take depends largely on the type of case you're investigating
   - Corporate, civil, or criminal
3. Refining and Modifying the Investigation Plan
   - Even if initial plan is sound, at times you may need to deviate from it and follow evidence
   - Knowing the types of data to look for helps you make the best use of your time
   - The key is to start with a plan but remain flexible in the face of new evidence

Analysis Methodology

1. Define and understand objectives
2. Obtain relevant data
3. Inspect the data content
4. Perform any necessary conversion or normalization
5. Select a method
6. Perform the analysis
7. Evaluate the results

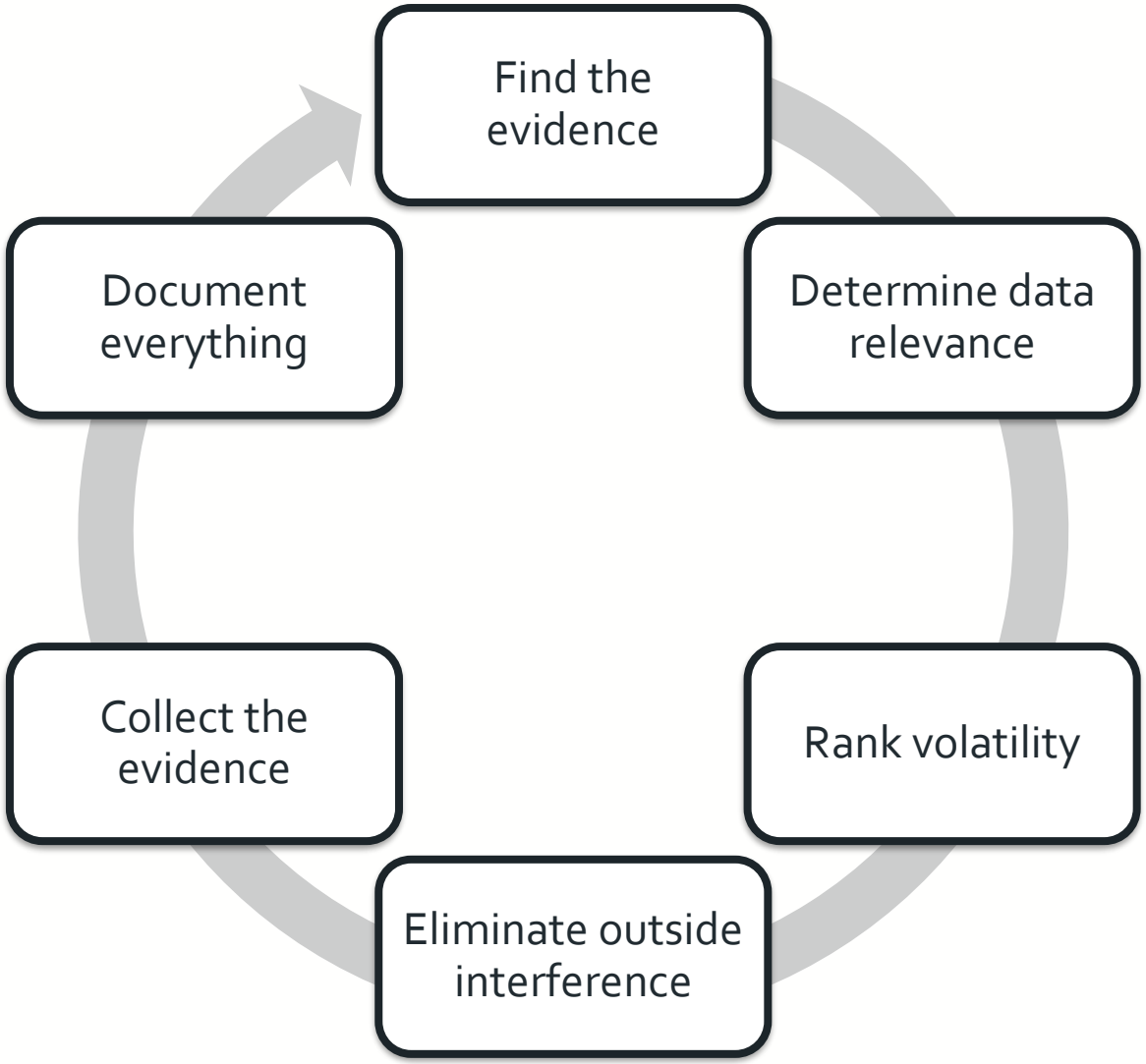
Forensic Examination in Practise



Phase 1. Collection

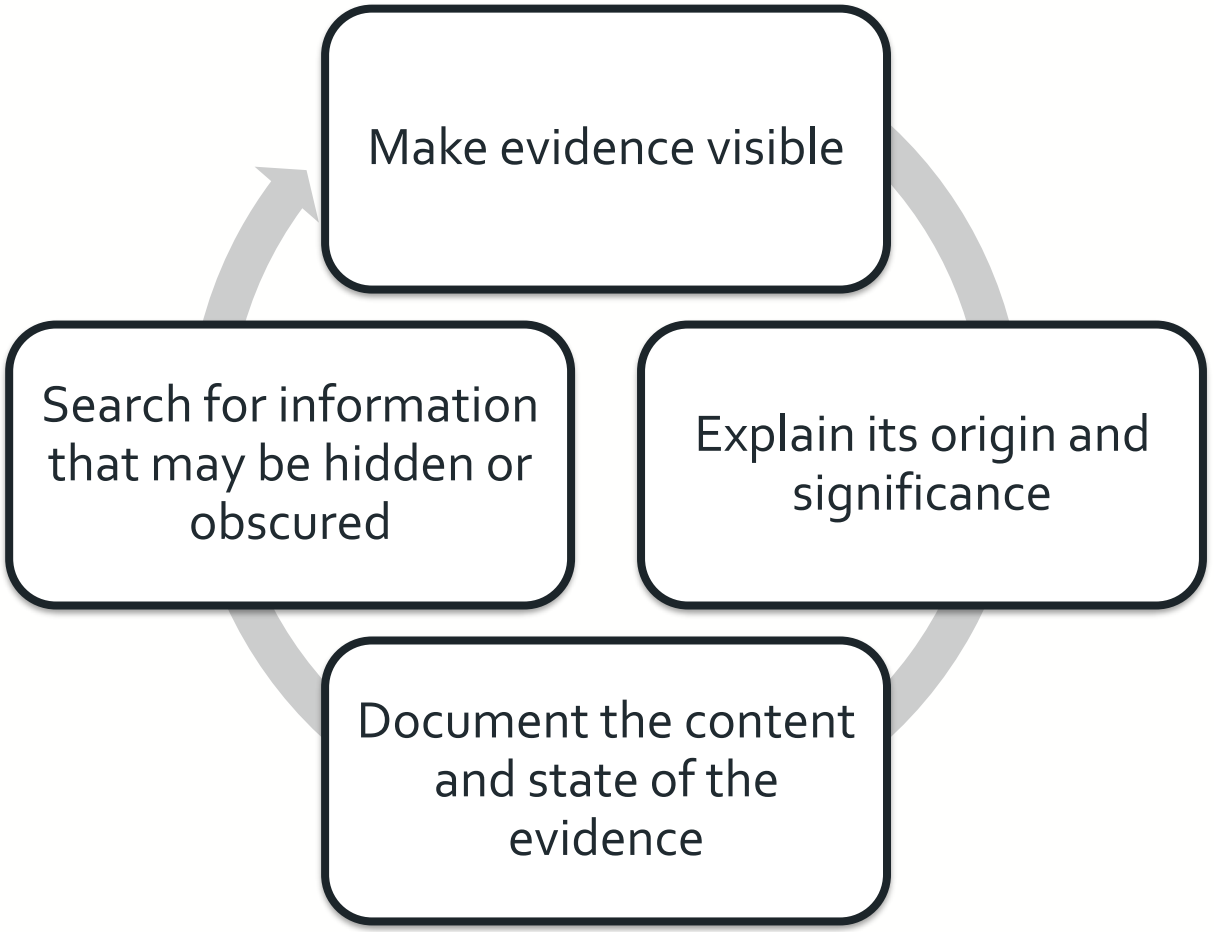
Artefacts considered to be of evidentiary value are identified and collected

- OOV
- Preservation
- Toolkit
- Chain of custody
- Admissibility

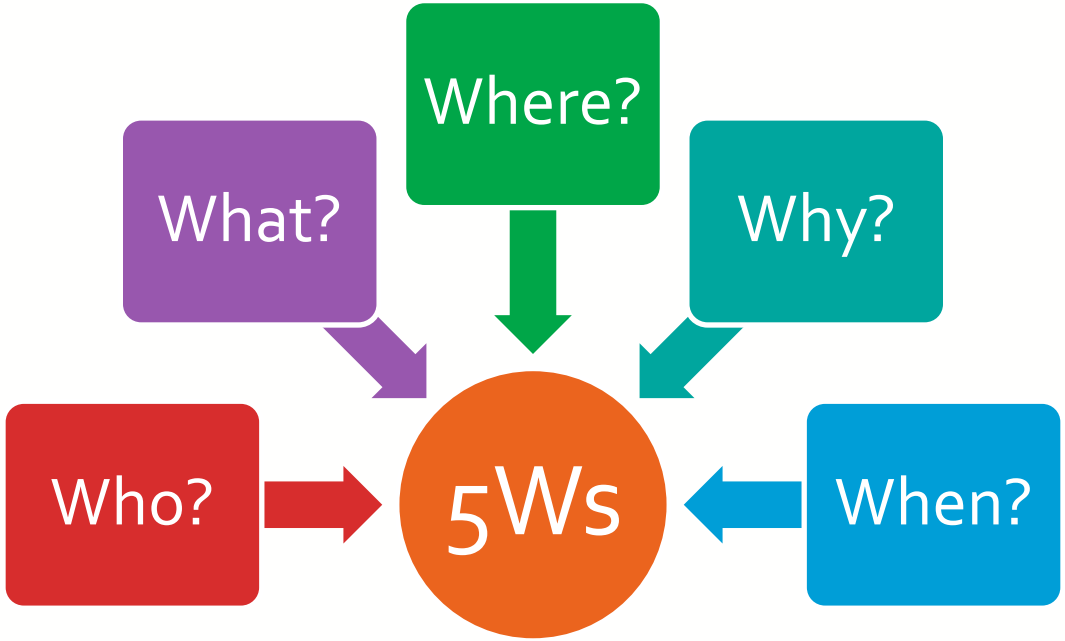


Phase 2. Examination

Examine the medium for evidential data



Phase 3. Analysis

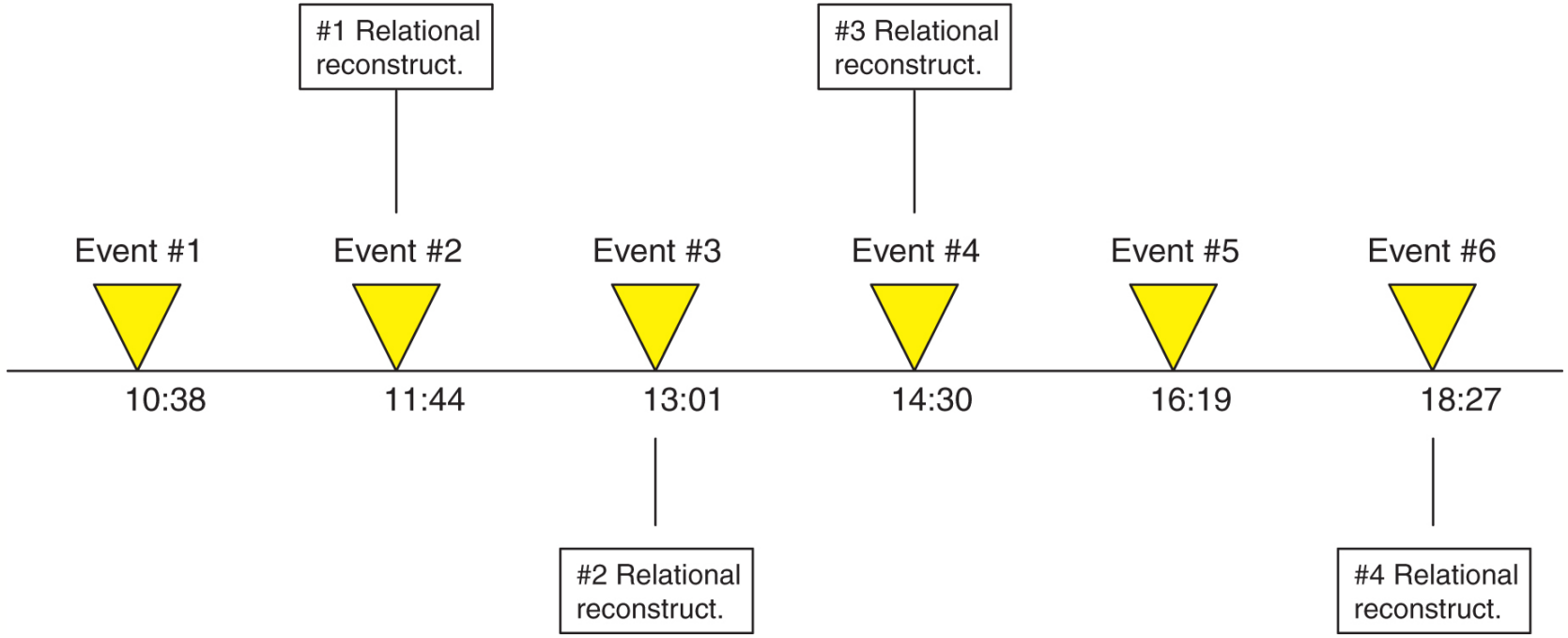


Filter data artefacts of evidential value

Answer to: **HOW?**

The timeline

- The outline of the incident
  - Aim is to identify digital timestamps
- A linear event analysis to establish
  - The sequence of events
  - Any related events



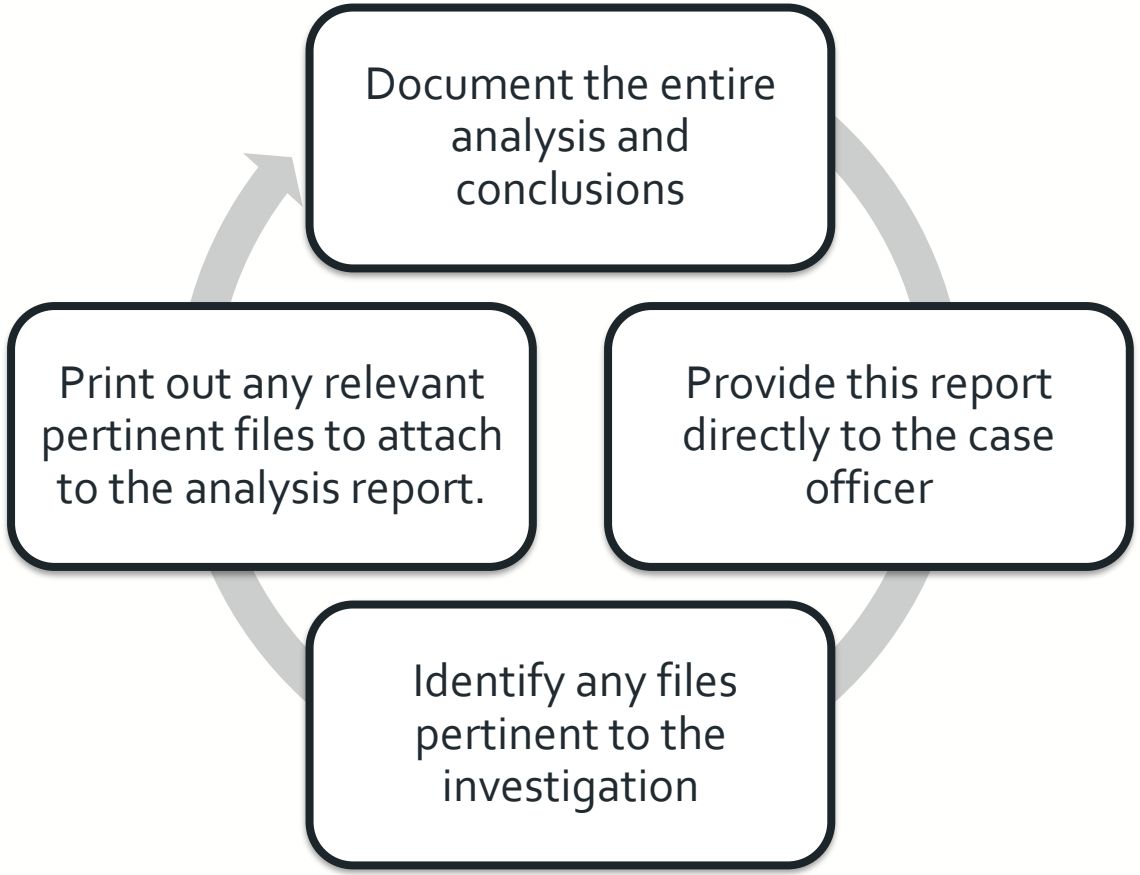
Phase 4. Report

Supporting documentation

Declarations

Statements

Testimony in court



Questions?

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References and Recommended Reading

- Casey, E. (2011). *Digital Evidence and Computer Crime: Forensic Science, Computers and the Internet*. Academic Press Inc (London) Ltd. ISBN 978-0-12-374268-1
- Nelson, B., Phillips, A., Steuart, C. (2018) Guide to Computer Forensics and Investigations (6th. ed.). Cengage Learning
- RFC 3227 - Guidelines for Evidence Collection and Archiving